From Lab to Market: What's Left of the Dye-Free Premium?

Price Signals in the Middle of America's Food-Color Transition

OPALS Summer Research Proposal · June 2026

Overview

In April 2025, HHS and the FDA announced the largest food-additive phase-out in American history: voluntary elimination of all six remaining petroleum-based synthetic dyes (Red 40, Yellow 5, Yellow 6, Blue 1, Blue 2, Green 3), with a voluntary compliance target of October 2027. Red No. 3 was formally revoked in January 2025 (enforcement: January 2027). As of June 2026, most major manufacturers have pledged reformulation, though the majority are still "In Progress" (FDA Industry Tracker, May 2026).

Before this shift, products marketed as "dye-free" or "no artificial colors" commanded price premiums. Meta-analyses estimated average WTP premiums of 30% for health and sustainability claims broadly (Li & Kallas 2021; Alsubhi et al. 2023) — but no published study has isolated the premium for "dye-free" specifically. As dye-free transitions from differentiator to expected baseline, the central question:

What remains of the dye-free price premium during this regulatory transition — and does claiming "no artificial colors" generate value independent of composition?

Early evidence suggests: very little. PepsiCo's Doritos Simply NKD retails at the same price as regular Doritos at Walmart (Food Institute, Feb 2026). Walmart's private-brand reformulation promises parity pricing (Walmart Corporate, Oct 2025). This makes mid-2026 a uniquely revealing measurement window.

OPALS Bridge. OPALS students in the food dye toxicity track study cellular effects of synthetic dyes — the same compounds the FDA is phasing out. This project traces the downstream impact: **Lab Finding → Regulatory Action → Industry Reformulation → Consumer Market Response** (Wang et al. 2025).

Novel Angle

Most WTP studies use stated-preference methods during stable regulatory periods. This study measures **observed retail prices** during an active voluntary phase-out — a regulatory situation without precedent in US food policy. The novelty is **situational**, not methodological.

No published study has isolated WTP for "dye-free" as a standalone attribute. This provides the first observed-price estimate for this specific attribute during the transition. We also test whether the **act of claiming** "no artificial colors" — independent of compositional difference — is associated with higher prices.

Theoretical Framework

1. Signaling Theory (Spence 1973). "No artificial dyes" is a voluntary quality signal. As adoption spreads, the signal's informational value decreases. *Prediction:* premiums lower where most competitors have adopted.

2. Anticipated-Mandate Expectations. The phase-out is legally voluntary as of June 2026 — no formal rule-making has revoked Red 40's authorization (PolicyCanary 2026; Haynes Boone 2025). But market participants price in the anticipated mandate. Firms reformulate pre-emptively; consumers discount the signal.

3. Competitive Price Convergence (Tirole 1988). Simultaneous industry reformulation prevents any firm from sustaining a premium — consumers substitute to identically-positioned competitors. PepsiCo's parity pricing for Doritos NKD illustrates this.

4. Cost Pass-Through. Observed price differentials reflect BOTH supply-side cost increases (natural colorants cost 2–10× more; per-SKU reformulation costs $50K–$200K per Haskell 2026) and demand-side WTP. We cannot fully disentangle these with observational data, but design specific comparisons to bound each component (RQ3).

The three demand-side mechanisms are **complementary, not competing —** all predict premium compression. The null-result framework (below) partially distinguishes them via data patterns.

Research Questions

#	Question	Prediction	Challenged If
RQ1	Do dye-free products command a measurable price premium over comparable dyed products in mid-2026?	Premium ≥ 5% in paired comparisons	Premium < 5% or non-significant. A null result is substantively meaningful — see *Interpreting Null Results.*
RQ2	Is the premium smaller where reformulation is more advanced?	Premium in high-reformulation categories ≤ half of low-reformulation	Premiums uniform across categories — would suggest cost/brand effects, not signal value. *Exploratory:* 25–30 pairs/category limits power.
RQ3	Among compositionally similar dye-free products, are those with “no artificial colors” claims priced higher?	Claimed > unclaimed	No difference — consumers respond to ingredients, not marketing. *Note:* association, not causal test (see Limitations).

Meta-analytic context (Discussion only): We compare findings against published WTP for broader health claims: 29.5% for sustainable food (Li & Kallas 2021) and 30.7% for healthier food (Alsubhi et al. 2023). These aggregate across organic, fair trade, reduced-fat — not “dye-free” specifically.

Regulatory Context (June 2026)

Date	Event	Legal Status
Jan 2025	FDA revokes Red No. 3	Formal revocation (Delaney Clause)
Apr 2025	HHS/FDA announce phase-out of remaining 6 dyes	Voluntary guidance (MAHA)
May 2025	FDA approves 3 new natural color additives	Formal approval
Feb 2026	FDA changes “no artificial colors” labeling — allows claim if no petroleum-based dyes, even with natural colorants	Enforcement discretion
Jan 2027	Red No. 3 enforcement date	Mandatory
Oct 2027	Compliance target for remaining 5 dyes	Voluntary — no formal rulemaking

> *“The FDA’s announcement does not include changes to underlying federal statutes, nor does it include the immediate revocation of all of the FDA’s regulations authorizing the impacted color additives.” — Haynes Boone, Apr 2025*

Industry Reformulation Status

Sources: FDA Industry Tracker (May 2026), CSPI Corporate Commitment Tracker (Apr 2026), PlainFoodSafe (Jun 2026).

Status	Companies
Complete	Sam’s Club (Member’s Mark), In-N-Out, Tyson, PepsiCo (NKD line)
In Progress — 2026	Hershey, Kraft Heinz, Danone, Grupo Bimbo, Campbell’s
In Progress — 2027	General Mills, Nestlé, Mars, WK Kellogg, Conagra, McCormick, Mondelez, PepsiCo full line, Walmart private brands, Coca-Cola, Post
No commitment	National Confectioners Association members (Ferrara/Ferrero, HARIBO, Tootsie Roll) — “will follow regulatory guidance” (CSPI)

Key: The National Confectioners Association has NOT pledged voluntary removal — confirming candy/confections as the right low-reformulation category for RQ2.

Product Classification

Based on composition and the February 2026 FDA labeling policy:

Group	Description	Example
A — Synthetic	Contains ≥1 FD&C petroleum-based colorant	Froot Loops (Red 40, Yellow 6, Blue 1)
B — Natural	No synthetic dyes; contains natural colorants. Can claim “no artificial colors” under Feb 2026 policy.	Annie’s Bunny Grahams (beet juice)

Group	Description	Example
C — None	No added colorants of any kind	Plain Cheerios

Data Sources

Source	Provides	Validation
USDA FDC — Branded Foods	Ingredient lists for >400K branded US products. Manufacturer-submitted, updated monthly. Search directly for "Red 40" etc.	Pre-camp: query for dye names, spot-check 30 products against Walmart.com, assess category coverage.
Open Food Facts	Supplementary ingredients + front-of-pack claims. European-origin, crowd-sourced.	Spot-check additive tag accuracy. Record `last_modified_t`. If >10% discrepancy or >50% pre-2025 → FDC primary, OFF claims only.
PlainFoodSafe.com	US product ingredient index (OFF data); dye product counts.	Verify usability. Counts may include non-US products.
Walmart.com	Current retail prices for matched pairs (manual lookup).	Pre-identify ≥85 candidate pairs. See Price Collection Protocol.
FDA + CSPI Trackers	Company-level reformulation pledges.	Cross-reference both. CSPI is more rigorous than FDA tracker.

Retailer note: Walmart's EDLP strategy compresses price variation, and Walmart itself pledges parity pricing (Walmart Corporate, Oct 2025). Griffith & Nesheim (2013): organic premiums 0% at discount retailers to 30% at premium retailers. **A positive premium at Walmart is a conservative lower bound; a null premium is ambiguous.** Findings are Walmart-specific, not market-wide. No scraping — manual lookup of publicly displayed prices only.

Methodology

Pre-Camp Preparation (Mentor, 25–30 hours)

#	Task	Details
1	**FDC data pull**	Query Branded Foods API for products with synthetic dye names. Filter to target categories.
2	**OFF data pull**	Download CSV, filter to US. Quantify additive-tagged products (E129/E102/E110/E133/E132/E143).
3	**Spot-check**	30 FDC + 30 OFF products verified against Walmart.com. Record discrepancy rate. For OFF: record `last_modified_t` distribution.
4	**Matched pairs**	≥85 candidates across 3 categories: (a) candy/confections 30 (low reform.), (b) cereals/baked goods 30 (med-high), (c) beverages 25 (medium). Document match type (within-brand / cross-brand), size (±25%). Target ≥15 within-brand.
5	**RQ3 products**	≥30 dye-free WITH "no artificial colors" claim + ≥30 WITHOUT. Same category, comparable brand tier.
6	**Geographic pilot**	5 products at zip 92093 vs. 77001. If identical → reduce geo check to 10 pairs.
7	**Templates**	Google Sheets with validation rules + 10 examples. JASP template with one example pair.
8	**Feasibility gate**	< 50 valid pairs → pivot to landscape-only study.
9	**Progress check**	2 weeks pre-camp: < 40 pairs → early gate. < 25 pairs → landscape only.

Software

Function	Tool
Data collection	Google Sheets (collaborative)
Statistical analysis	JASP (free, GUI; jasp-stats.org) — supports all required tests
Visualization	JASP + Google Sheets charts
Poster	Google Slides or PowerPoint

Price Collection Protocol

1. All prices from **Walmart.com**, primary zip **92093** (UCSD)
2. **Geographic check:** 10–20 pairs at zip **77001** (Houston, TX) per pilot results
3. Collected within a **3-day window** (Week 1, Days 3–5)
4. Record **regular shelf price:**
   - No badge/strikethrough → displayed price is regular
   - "Rollback" WITH strikethrough → record both; use strikethrough for primary analysis
   - "Rollback" WITHOUT strikethrough → displayed price (permanent rollback)
   - "Was $X.XX" → record both; use "Was" as regular
   - Do NOT use Walmart+ member pricing
5. Record **date of collection** per product
6. **Screenshot every 10th product** as audit trail
7. **Inter-collector reliability:** 10% overlap subsample by two students. Discrepancy >5% → reconcile

Week 1: Data Assembly & Familiarization (5 days)

Days 1–2 — Learning:
- "Ingredient detective" exercise: classify 10 familiar products into Groups A/B/C
- FD&C naming conventions, E-numbers, natural colorant names
- Regulatory timeline and February 2026 labeling change
- "Price pair" demo: look up 3 pre-selected pairs, record all fields, discuss

Days 3–5 — Data collection. Record per product: name, brand, category, dye status (A/B/C), dye count, front-of-pack claims, brand tier (premium/mainstream/value), price, package size (total oz), unit price (price/oz solids, price/fl oz liquids), match type (within-brand / cross-brand).

End of Week 1 go/no-go:
- ≥60 valid pairs across ≥2 categories → proceed to RQ1/RQ2
- ≥15 within-brand pairs → within-brand premium is primary estimate
- < 15 within-brand → proceed with caveat (cross-brand carries brand-equity confounds)
- < 60 valid pairs → drop pricing; focus on reformulation landscape

Week 2: Statistical Analysis (5 days)

RQ1 — Primary. Paired Wilcoxon signed-rank test. Report median premium %, IQR, p-value. **Stratify by match type:** within-brand pairs are the clean comparison — if within-brand ≈ 0% but cross-brand > 0%, interpret the latter as brand positioning. Package-size check: sensitivity analysis restricted to pairs within ±25%.

RQ2 — Exploratory. Kruskal-Wallis across three categories; pairwise Mann-Whitney U with Bonferroni (3 comparisons). 25–30 pairs per category. Report effect sizes (rank-biserial r) alongside p-values.

RQ3 — Exploratory. Mann-Whitney U: claimed vs. unclaimed dye-free unit prices within category. Sensitivity: restrict to same-brand-tier. 30 per group; power 0.50–0.60 for small effects.

Week 3: Reformulation Landscape + Review Coding (5 days)

Phase 3 — Reformulation Landscape (all students, primary deliverable)

Using FDC + OFF + FDA Tracker + CSPI Tracker:
- **Dye prevalence:** % of products per category still containing synthetic dyes; which dyes most prevalent
- **Dye complexity:** distribution of distinct dyes per product by category (3+ dyes = higher reformulation cost)
- **Pledge-vs-product-data gap analysis:** cross-reference company pledges (FDA/CSPI) with product-level data (FDC primary). Have companies that pledged by mid-2026 actually removed dyes?
- **Staleness validation:** manually verify ≥10 apparent gaps against Walmart.com. Report confirmed gaps vs. database lag.
- **State regulatory exposure:** which products with dyes are sold in states with bans? CA (AB 418, 2023); WV (school meals, 2025); >70 bills, >15 enacted (EWG, Mar 2026)
- **Visualize:** stacked bars by category; compliance timeline; dye heat map; pledge-vs-reality chart

This analysis is independently publishable regardless of pricing outcomes. No existing tracker cross-references company pledges with product-level ingredient data.

Phase 4 — Review Content Analysis (stretch goal)
30–40 products + 5–10 neutral controls. 10 most recent reviews per product, ≥2 sentences (manual reading only). Code: dye/health mentions, taste, switching behavior, price sensitivity. Inter-coder reliability: 20% overlap, Cohen's $\kappa \geq 0.6$.

Week 4: Synthesis & Communication (5 days)

- Integrate: landscape × premium estimates × review themes
- IMRAD-format poster + oral presentations
- **Stretch:** 2-page conference abstract

Interpreting Null or Near-Zero Premium Results

If observed premium < 5% or non-significant:

Interpretation	Mechanism	Supporting Data Pattern
Transition completion	Market has priced in the baseline shift	Premium ≈ 0% in high-reform. categories, > 0% in low-reform.
Competitive convergence	Simultaneous reformulation prevents any firm sustaining a premium	Premium ≈ 0% uniformly across all categories
Cost absorption	Firms absorb higher costs to maintain share	Premium ≈ 0% for major brands, > 0% for small brands
Attribute non-salience	Dye-free was never independently valued	Premium ≈ 0% overall, but RQ3 claim association positive
Attribute inversion	Dye-free is mainstream; dyed products are niche/legacy	Negative premium, driven by cross-brand; within-brand ≈ 0%

A null result is not a failed study. Documenting the absence of a premium during this regulatory moment establishes an observed-price baseline and tests theoretical predictions about premium erosion during industry transitions.

Deliverable Tiers

Tier	Deliverable	Depends On
Must ship	Reformulation landscape: dye prevalence, pledge-vs-product-data gap analysis with staleness validation, dye-complexity scoring, state exposure. + poster + presentation	FDC/OFF data only
Should ship	Matched-pair premium estimates for ≥60 pairs across 3 categories (RQ1, RQ2)	Price collection succeeds
Should ship	Claim-vs-no-claim comparison (RQ3) with brand-tier sensitivity	≥30 products per group
Stretch	Review content analysis with inter-coder reliability	Time and capacity
Stretch	Conference abstract with meta-analytic context	Statistical sophistication

Scope Boundaries

- Takes a **cross-sectional snapshot.** No longitudinal claims.
- Reports **descriptive/correlational** premiums. No causal identification.
- Treats RQ3 as an **association**, not a causal test.
- Reports **Walmart-specific** findings, not market-wide estimates.
- Does not study PFAS, Reddit, or use scraping/API-based price collection.
- Does not assume a positive premium. Null and negative findings are substantively meaningful.

Limitations

1. **Cost pass-through confound.** Observed differentials include manufacturer cost increases (natural colorants 2–10× more; Haskell 2026). Cannot fully disentangle supply and demand. RQ3 partially addresses.
2. **Selection bias.** Within-brand pairs exist disproportionately for brands not yet reformulated — a non-random sample.
3. **Ingredient data coverage.** FDC is more systematic than OFF for US products but may not cover all items. Landscape findings describe products in the database, not all US products.

4. **Single time point.** Cannot measure change over time. Meta-analytic comparisons are suggestive context, not formal benchmarks.
5. **Voluntary transition.** Red 40 remains lawfully authorized. We study an anticipated-mandate, not post-mandate environment.
6. **RQ3 endogeneity.** Claiming brands are systematically premium-positioned. Controls for composition, not brand positioning or bundled claims.
7. **RQ3 power.** 30/group; power 0.50–0.60 for small effects. Exploratory; non-significance may reflect power.
8. **Single retailer, single geography.** Walmart EDLP compresses variation; California is the most regulated state for food dyes. Geographic check partially addresses. Griffith & Nesheim (2013): 0–30% organic premium range across retailer types.
9. **Database staleness.** Even FDC has update lags. Staleness validation (≥10 manual checks) partially addresses.

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